

Scoped Falsification Map for White-Box Neural Integration

Algorithmic-contribution closeout article — 30 July 2026

Abstract

We report a reproducibly labeled campaign of finite-width white-box correction experiments around a strong 66,048-point Kerdock baseline. The goal is not to prove universal impossibility, but to identify which tested information classes fail complete deployment after accounting for signed alignment, tail risk, selection chronology, reference noise and compute. Several legal methods carry real average signal, and per-network oracle corrections can be large. However, the tested fixed policies, scalar predictors, harmonic dictionaries, analytic anchors, Poisson controls, projected-ReLU controls, coresets and signed near-collision probes fail because their signed phase is unstable, their gains disappear under grouped frozen extension, their tails are unsafe, or their compute cost exceeds the benefit. These results define a falsification map and explicit reopening conditions; they do not rule out new observables, arbitrary nonlinear estimators or new signed geometry.

1. Evidence labels

Every result must carry one of four labels:

- **FROZEN EMPIRICAL** — selection frozen before a grouped evaluation set;
- **ORACLE DIAGNOSTIC** — uses unavailable target information or per-instance fitted coefficients;
- **EXPLORATORY** — post-selected or insufficiently separated;
- **PROVISIONAL / MISSING** — primary package cannot be reconstructed.

Oracle and deployable numbers must appear in separate tables and figures.

2. Correct performance decomposition

For baseline error e and correction c ,

$$\mathbb{E}\|e - c\|^2 = \mathbb{E}\|e\|^2 - 2\mathbb{E}\langle e, c \rangle + \mathbb{E}\|c\|^2.$$

The correct question is signed, downstream-weighted alignment, not whether a latent state is accurately predicted in an unweighted Euclidean norm.

For runtime information G , the best unrestricted G -measurable correction is

$$\mathbb{E}[e \mid G],$$

and the information value is

$$V(G; e) = \mathbb{E}\|\mathbb{E}[e \mid G]\|^2.$$

A large target-labeled oracle span does not imply that the legal runtime information contains the oracle phase.

3. Target-noise limitation

Suppose an observed learning target is

$$Y = \theta + \varepsilon,$$

with mean-zero reference noise independent of features and the latent target. Then

$$R_Y^2 = \frac{\text{Var}(\mathbb{E}[\theta \mid X])}{\text{Var}(Y)} = \frac{\text{Var}(\theta)}{\text{Var}(Y)} R_\theta^2.$$

Thus observed predictability is attenuated by target reliability. Negative or weak grouped R^2 against a noisy reference cannot establish zero latent predictability. The Path-2 result is therefore retained only as a matched-model comparison in its specific information class.

4. Main empirical map

- **T4 fixed policy — frozen development failure.** Raw ratio 1.127854; failure mode: low signed alignment and severe tails. Reopen only with a new feature class frozen before grouped validation.
- **Small degree-6+8 dictionary — frozen failure.** Raw ratio 1.004439; no stable gain. Reopen only with a preregistered richer live-degree family and an oracle ceiling.
- **Path-2 scalar models — narrow negative.** No value over matched constants. Reopen only with reliable targets and new noninvariant features.
- **Archived downstream anchors — diagnostic, not submission-ready.** Pooled raw 0.851978, worst 1.924; rotation-dependent phase and tails. Reopen only with a stable downstream-weighted phase predictor and tail guard.
- **Poisson controls — tested dictionary closed.** Exploratory 0.929, terminal 1.038; post-selection did not transfer.
- **Projected ReLU controls — tested family closed.** Sixteen-network 0.927, 48-network 1.013, adjusted 1.067; hard-network fluctuation and cost.
- **Outside-Kerdock signed probes — phase benchmark.** Oracle 0.131, frozen global 1.557; enormous capacity but nontransferable coefficients.
- **M146 — provisional/non-evidence.** Primary artifacts are missing and the scalar gate is not invariant.
- **M152 — removed.** No primary package survives.

5. Reopened paths

5.1 Downstream-weighted anchor rescoring

The rescoring campaign confirms that unweighted latent-state precision can discard useful output directions. The best frozen external arm reaches pooled raw ratio 0.851978 across 28 external cases, but p90 is approximately 1.341, worst case 1.924, and the rotation-expansion subgroup reverses above one. This is meaningful diagnostic evidence and an unsafe deployable estimator.

5.2 Network-derived Poisson controls

A post-hoc exploratory mid-radius dictionary reaches approximately 0.929, but the frozen terminal result is 1.038. The tested directions and radii are closed; the existence of analytically integrable high-degree controls remains open.

5.3 Exact-mean projected ReLU controls

A candidate initially reaches 0.927 on 16 terminal networks. Without refitting, a 48-network extension reverses to 1.013; even a favorable implementation model gives adjusted ratio 1.067. This branch fails both statistical and economic gates.

5.4 Outside-support signed probes

Per-network target-labeled coefficients can reduce pooled error to approximately 0.131, while one frozen global coefficient vector scores 1.557 with catastrophic tails. The correct conclusion is not “signed probes cannot work,” but “the tested weight-derived representation does not transport their signed phase.”

5.5 Final exact-mean and anchor descendants

A final preregistered campaign tested high-degree exact-mean sphere-Stein controls and a downstream-weighted T4 shrink/abstention descendant. The primary Stein rule scored candidate/base 1.010720 on 16 fresh networks, with nine wins and worst ratio 1.13020. A downstream-singular-vector-aligned descendant showed only about a 1.3% development gain with an interval touching no gain and did not justify its Jacobian/SVD cost. The signed-calibration descendant used negligible negative mass and inherited the failed validation result.

The fresh T4 shrink policy scored candidate/base 1.000097 on eight new base networks and three literal rotations each, with only three of eight network wins and worst network 1.06802. The same correction span retained a target-labeled oracle gain of 1.167584x. This is the cleanest final demonstration of the capacity/observability distinction: the span contains useful directions, but the tested runtime features do not identify their signed phase.

5.6 Frozen implementation closeout

The seven immutable shipping packages were run as isolated local subprocesses on the deterministic synthetic contract. A42 and A43 were bit-identical and materially reduced local wall time and peak memory relative to the production package; A42 was fastest in this one run. Because the official grader, FlopScope installation, official cohort and raw-MSE/effective-cost calculation were unavailable, the local comparison cannot promote a package. The production package remains the externally verified shipping choice pending an official-environment run.

6. Group-invariant observability

If runtime diagnostics are invariant under a measure-preserving orientation reversal while the candidate correction direction changes sign, their conditional signed alignment is zero under the symmetry model. Norms, Gram matrices, disagreement magnitudes and condition numbers can then predict capacity without predicting orientation. The actual WHestBench distribution has not been proved exactly symmetric; grouped cross-fitting should estimate the size of any invariant component rather than assume it vanishes.

7. Contamination and grouping rules

- Rotations of one base network are not independent observations.
- Every confidence interval and split must group by base network.
- Exposed Mini-100 or development panels are development only.
- Truth-stream construction and noise correction must be stated per row.
- Post-hoc descendants require a new terminal set.
- A frozen extension without refitting is stronger evidence than another small fresh split selected after inspection.

The accompanying `EMPIRICAL_EVIDENCE_AND_CONTAMINATION.csv` is the canonical chronology table.

8. Compute and tail safety

A candidate must report raw MSE, grouped wins, p90, worst case, reference uncertainty, incremental FLOPs, residual wall time and complete adjusted score. Mean gain alone is insufficient. The reopened campaign demonstrates that methods with positive pooled signal can still be unusable because the tails are larger than the mean benefit.

9. Claims removed

The article must not say:

- no statistical path exists;
- all legal corrections have zero alignment;
- scalar learning is impossible;
- the anchor tolerance is universally approximately $5e-4$;
- analytically integrable controls are low degree;
- degree-6+ controls cannot help;
- signed weights are globally closed;
- infinite-width harmonic shares are width-256 measurements.

10. Reopening map

Further work requires at least one genuinely new ingredient:

1. a legal absolute-phase observable not contained in the tested invariant feature classes;
2. an analytic per-network coefficient for off-support signed probes;
3. a live-degree exactly integrable surrogate with a preregistered oracle ceiling and residual-kernel recertification;
4. finite-width-specific arbitrary-node geometry;
5. a target with substantially higher reliability for learning studies.

11. Project stopping rule

No active branch in the tested information classes clears a credible continuation gate under the current evidence, deadline and resource constraints. A branch should reopen only with a genuinely new runtime information source, geometry or residual transformation that is frozen before protected evaluation and demonstrates either at least $1.3\times$ raw improvement with safe grouped tails or a complete measured adjusted-score win.

12. Conclusion

The experiments show a repeated separation between capacity and observability. Strong oracle directions and average legal signal exist, but the tested methods fail complete deployment because signed phase is unstable, reference labels are noisy, tails are unsafe, or compute dominates. This is a useful and reproducible negative map. It is not a universal no-free-lunch theorem.